



Multimodal IRA

Gemma 4 Multimodal Companion / Roleplay Model Assignment

Context

Ira is a companion AI product, not a generic assistant. The goal is to build a strong Gemma multimodal-based companion / roleplay model that can approach or exceed the current experience on a defined set of behaviors.

You can assume:

- full GPU access will be provided
- you are allowed to run training jobs
- multimodal support is expected

Objective

Take a Gemma multimodal base model and turn it into a high-quality companion / roleplay model for Ira using:

- SFT
- RL / preference optimization
- good data design
- strong evaluation
- deployable inference

The final output should not just be "a fine-tuned model." It should be a model that feels:

- warm
- emotionally aware
- consistent in character
- good at roleplay
- less like a generic assistant
- suitable for real product testing

A core requirement is strong multilingual behavior, especially for:

- Hinglish
- Latin-script Indic languages
- code-switched conversations common in real messaging behavior

Core Task

Build a complete training + evaluation + inference pipeline for a Gemma-based companion model.

Your work must include:

- base model selection and justification
- supervised fine-tuning
- RL / preference optimization stage
- roleplay / companion-specific evals
- multilingual evaluation
- deployable serving stack
- final recommendation on readiness

Scope

The system should support:

- multi-turn conversation
- text input
- image input
- system / persona prompting
- memory-context injection
- controllable response style

- API-based inference
- multilingual conversation quality for Hinglish and Latin-script Indic use cases

If you choose to support audio input too, explain the architecture clearly, but the core focus is companion / roleplay quality.

Required Workstream 1: Model Strategy

Choose the Gemma variant you believe is best suited for this use case.

Your write-up must explain:

- why this model was chosen
- what tradeoffs you are making on quality, speed, and memory
- whether you are using full fine-tuning, LoRA, QLoRA, or another approach
- what multimodal capabilities are used in practice
- how you expect the model to handle Hinglish and Latin-script Indic conversations

Required Workstream 2: SFT

Build an SFT pipeline that meaningfully improves companion behavior.

This must include:

- training data schema
- data curation / filtering approach
- train / validation split
- reproducible configs
- at least one strong SFT checkpoint
- training logs and hyperparameters

The SFT stage should improve:

- tone
- roleplay adherence
- warmth
- conversational naturalness
- reduction of generic assistant language
- multilingual fluency in Hinglish and Latin-script Indic chat

Your data plan should explicitly address:

- code-switching behavior
- transliterated Indic text
- informal messaging style
- slang / mixed-language expression
- multilingual emotional expression

Required Workstream 3: RL / Preference Optimization

After SFT, run an alignment stage focused on roleplay / companion quality.

Acceptable approaches:

- DPO
- ORPO
- PPO or another RL method, if justified clearly

This stage should target:

- emotional nuance
- character consistency
- memory-faithful behavior
- reduced robotic phrasing
- better response quality in ambiguous or emotionally loaded turns
- better multilingual and code-switched conversational quality

You should explain:

- why you chose the method
- what preference data you used
- what changed from SFT to the aligned model
- whether alignment improved or degraded multilingual behavior

Required Workstream 4: Inference Service

Build a runnable inference service for the final model.

It must support:

- persona / system prompt injection
- memory-context injection

- multi-turn chat
- configurable decoding parameters
- structured logs for latency, throughput, and GPU memory

Provide:

- setup instructions
- run instructions
- example requests
- notes on deployment assumptions

The service should be able to handle:

- English
- Hinglish
- Latin-script Indic inputs
- mixed-language chat turns

Required Workstream 5: Evaluation

Build a companion / roleplay-specific evaluation suite.

The evals must cover:

- emotional sensitivity
- warmth / naturalness
- roleplay consistency
- memory faithfulness
- multimodal grounding
- playfulness / banter
- safety / boundaries
- recovery from ambiguous messages
- multilingual fluency
- code-switching quality
- transliterated Indic robustness

Example scenarios:

- "I'm fine" when the user is clearly not fine
- callback to an earlier personal detail

- image-sharing with a personal response
- playful banter without generic assistant tone
- emotionally dependent user behavior requiring warmth plus boundaries
- staying in-character over long conversations
- Hinglish chat that switches naturally between English and Hindi
- Latin-script Indic input with spelling variation and informal phrasing
- mixed-language emotional support conversations
- code-switched flirt / banter / comfort / casual chat

You should compare:

- base Gemma
- SFT Gemma
- RL / preference-tuned Gemma

If useful, include comparison against the current Gemini behavior.

Multilingual Requirement

This project must explicitly optimize for multilingual companion behavior.

At minimum, evaluate and improve for:

- English
- Hinglish
- Latin-script Hindi
- other Latin-script Indic variants if relevant to your approach

We care about:

- natural code-switching
- non-awkward transliterated responses
- correct tone in mixed-language chat
- informal messaging fluency
- emotional nuance across language mixing
- avoiding sterile or over-formal phrasing

The model should not simply “understand” these inputs. It should respond in a way that feels native to real user chat behavior.

Deliverables

Please submit:

- [README.md](#)
- `serve/`
- `training/`
- `preference/`
- `evals/`
- `results/`
- `final_report.md`

`final_report.md` must answer

- what model you chose and why
- what data you used
- how multilingual data was handled
- what SFT improved
- what RL / preference optimization improved
- where the model still fails
- whether the model is ready for:
 - narrow deployment
 - hybrid routing
 - broader rollout
- what the next highest-leverage improvement is

Success Criteria

Minimum success:

- reproducible setup
- one meaningful SFT checkpoint
- one meaningful RL / preference-tuned checkpoint
- evaluation suite that reflects the actual product problem
- explicit multilingual evaluation
- honest analysis of failure modes

Strong success:

- clear gains from base -> SFT -> aligned model
- meaningful improvement in tone, warmth, and roleplay quality
- reduced generic assistant behavior
- strong multilingual behavior in Hinglish and Latin-script Indic chat
- model is good enough for internal product testing

Exceptional success:

- model is strong enough for a real slice of traffic
- quality is high enough that deployment is a realistic next step
- multilingual/code-switched behavior feels natural and product-ready
- recommendation is grounded in evidence, not optimism

Non-Goals

Do not:

- optimize mostly for generic benchmarks
- confuse long context with memory
- overfocus on infra while under-solving behavior quality
- claim success based on vague qualitative impressions
- skip failure analysis
- stop at SFT if the model still feels generic
- ignore multilingual quality while optimizing English-only performance

What Matters Most

We care most about:

- quality of data judgment
- quality of eval design
- ability to shape model behavior
- multilingual and code-switched fluency
- ability to make the model feel human without becoming messy
- ability to preserve warmth, consistency, and boundaries together
- ability to make a clear product recommendation at the end

Final Decision Output

Your final recommendation must end with one of:

- ready for narrow deployment
- ready for hybrid routing
- not ready yet